

AMRUTHA SOMASHEKHAR

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Adept Data Analyst with over **4 years of expertise** in data analytics, ETL development, business intelligence, and cloud computing across finance and technology sectors. Expert in data integration, advanced statistical modeling, and predictive analytics using SQL, Python, and AWS technologies. Proven ability to design data pipelines, develop interactive dashboards, and ensure data integrity and compliance. Skilled in cross-functional collaboration, data governance, and delivering actionable insights that drive business decisions.

Professional Experience

Data Analyst, Bank of America, Pennington, New Jersey

August 2024 – Present

- Designed and optimized SQL-based data pipelines, enhancing data integration and processing efficiency by 30%, ensuring high data integrity for regulatory reporting and seamless data flow.
- Developed interactive Tableau and AWS QuickSight dashboards, visualizing risk exposure, credit risk trends, and operational efficiency metrics, enabling data-driven decision-making.
- Automated data validation and anomaly detection workflows leveraging Python-based AI models, significantly reducing manual efforts and improving data quality by 35%.
- Built predictive analytics models to track financial risk patterns and enhance fraud detection capabilities using SQL and AWS QuickSight. Led cross-functional collaboration with risk management, compliance, and IT teams to streamline data governance policies, ensuring adherence to regulatory requirements.
- Created automated risk assessment dashboards in Tableau, reducing manual report generation by 50% and enhancing real-time decision-making capabilities.

Data Analyst, Tata Consultancy Services, Bangalore, India

July 2019- July 2022

- Orchestrated the extraction and transformation of 10M+ records utilizing advanced SQL queries and Excel functions, facilitating critical insights for strategic decision-making processes.
- Demonstrated strong analytical skills by extracting, collecting, organizing, analyzing, and interpreting complex data sets to identify trends and patterns, leading to a 22% reduction in operational inefficiencies.
- Optimized ETL workflows, achieving a 20% increase in efficiency and cost reduction through implementation of AWS cloud infrastructure, including EC2, S3, Lambda, Redshift, and QuickSight.
- Produced interactive Power BI dashboards to monitor 20+ KPIs, facilitating real-time business analysis and performance tracking.
- Implemented feature engineering techniques including intersection, normalization, and label encoding using Scikit-learn preprocessing, resulting in a 15% improvement in model accuracy and a weekly reduction of 20 hours in training time.

Professional Certification

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| Google Data Analytics Link | October 2024 |
| Python for Data Science & AI- IBM Link | February 2024 |
| AWS Certified Solution Architect – Associate Link | December 2023 |
| Applied Machine Learning Graduate Certificate | May 2023 |

Projects

AI-Powered Data Quality Automation for ETL Pipelines (Python, SQL, BigQuery, Snowflake, Tableau)

January 2024

- Developed an AI-powered anomaly detection framework for large-scale ETL pipelines, improving data integrity by 70% and implemented self-healing pipelines using Apache Airflow, reducing data failure rates from 15% to 1%.
- Automated schema evolution testing and CI/CD data validation, improving pipeline deployment efficiency and reduced manual data validation workload by 70% through ML-driven data quality automation.
- Employed AWS CloudWatch for real-time monitoring and utilized Amazon Connect for automated alerts, ensuring robust pipeline performance and heightened system responsiveness.

Intelligent Data Consistency and Drift Monitoring System (Python, SQL, Snowflake, Power BI)

December 2023

- Built an event-driven data drift detection system using Kafka and Apache Flink, enabling real-time anomaly tracking and reducing undetected schema mismatches by 40%.
- Implemented an automated Snowflake-based data reconciliation framework, ensuring data consistency across ingestion layers and improving data integrity by 35%. Leveraged advanced SQL and Python for data extraction and mining, integrating Alteryx to streamline data processing and apply advanced statistical techniques.

Technical Skills

- Data Analysis & Visualization: Advanced SQL, Tableau, Power BI, Alteryx, AWS QuickSight
- Programming Languages: SQL, PL/SQL, NoSQL, Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, ggplot2), R
- Database: MySQL, PostgreSQL, MongoDB, Oracle, BigQuery
- Cloud: AWS (EC2, Lambda, S3, IAM, Redshift, Sage Maker, CloudWatch), Snowflake, GCP, Azure
- Machine Learning - Linear Regression, Logistic Regression, Decision Trees, SVM, Random Forest, Scikit-Learn, K-Means Clustering
- Other Tools: Git, Snowflake, PySpark, Databricks, Microsoft

Education

The University of Texas at Dallas, Richardson, TX

May 2024

Master of Science, Business Analytics (Data Science concentration)

Relevant Courses: Big Data, Database Foundations for Business Analytics, Business Data Warehousing, Advanced Statistics for Data Science, Business Analytics with R, Predictive Analytics, Prescriptive Analytics, Applied Machine Learning, Applied Deep Learning, AWS Cloud Solution Architecture and Applied Econometrics and Time Series Analysis